

Dr.supER: Intubation and Ventilator Troubleshooting VR Simulation

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ABSTRACT

This paper presents Virtual Reality 3D interaction and simulation techniques in the context of the 2021 3DUI contest of the IEEE VR conference. In our implementation, by taking into consideration the current COVID-19 pandemic situation, we developed a VR gamified simulation regarding the intubation and ventilator troubleshooting procedures, taking place in an Emergency Room (E.R.) In this paper, we present the design and development of the overall experience, aiming to train the end-user on these actions to enhance their possibilities of achieving optimal performance under stressful situations. By gamifying the above education system, we also expect to empathize the layman regarding the daily challenges that the medical staff is experiencing in an Intensive Care Unit.

Index Terms: Ventilator, Intubation, Troubleshooting, Covid19, Virtual Reality, Gamification, Haptic Feedback, Mixed Reality

1 INTRODUCTION

State of the art technological feats such as virtual reality applications can enhance our teaching methods, in several ways due to their nature and their ability to provide novel-immersive experiences to end-users. Consequently, a significant amount of research aims on improving the virtual reality user experience with realistic graphics and practical 3D interfaces. Despite the great room for improvement, immersive reality applications pose as a promising tool to aid students and individuals, in grasping challenging notions and processes. [1]

This year, the IEEE VR 2021 holds the 12th annual 3DUI Contest, aimed to stimulate innovative and creative solutions to challenging 3DUI problems. The theme of this year is “Challenging Pandemics” and focuses on developing experiences related to the current COVID-19 situation. This may include VR/AR/MR driven 3DUI systems for training, teaching, treatment, or therapy support.

Taking into consideration the potential usability of immersive VR simulations in healthcare education and the increasing number of populations that needs to be hospitalized in Intensive Care Units (ICUs) [2], the proposed simulation focuses on empathizing the user on the real-life difficulties’ doctors face during the pandemic and especially on training medical staff on intubation procedures and ventilator troubleshooting techniques they may come across. Regarding the medical education and, more

precisely, the intubation of a patient, significant research took place in both previous conferences [3] and overall literature [4] which manages to integrate the haptic aspect to it. Our implementation, while sharing common features with previous work around the intubation procedures, contributes to the state-of-the-art literature by incorporating the ventilator troubleshooting aspect and promising effective training due to its gamified nature.

2 DESIGN

Owing to the COVID-19 pandemic, more than 0.5% of the world population needs to be hospitalized in ICUs. [5]. The cost of these units has the greatest share of overall medical costs. They usually require highly experienced and trained doctors not only for intubation procedures but also for the general monitoring of the patient when complications arise. Due to COVID-19’s high transmissibility, there is a need to educate medical personnel in a more specialized manner, to reinforce and decongest the medical system. The gamification of the overall experience could aid in making the user more knowledgeable and confident to perform in an optimal level under pressure.

In our simulation, there are three different available modes, two of which are closely related with the learning procedure (tutorials) and one (main game), where the previous acquired knowledge must be utilized to deal with any challenge that may arise in a VR Emergency Room. Briefly, the overall procedure begins with the emplacement of the patient in the correct head position to assist the laryngoscope and the endotracheal tube to penetrate the neck. Subsequently, laryngoscope enters the oral cavity to establish a clear view of the epiglottis and the cords as it shown in figure 1. Then the endotracheal tube passes through the trachea and the cords. A small balloon is attached to the tube’s head, which is being inflated after the tube is placed correctly to prevent any unwanted movement. Finally, we remove the laryngoscope and the ventilation starts [6]. These steps must take place in a 30 second period to prevent permanent deficiencies, and in 2-3 minutes to avoid fatal complications. Consequently, the respective training session may be able to enhance the therapist’s ability to properly implement crucial steps such as forcing a right body position, holding the blade correctly or efficiently protecting the oral cavity from injuries frequently occurring in the process.



Figure 1: The first and second step of intubation procedure.

Next, we proceed on illustrating the ventilator troubleshooting training simulation. To elaborate, every intubated patient has a

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private monitor that displays the graphs of the peak inspiratory pressure, the air flow, and the volume of the lungs during inspiration and expiration as represented in figure 2. There are numerous problems that may disrupt the collaboration of the patient with the ventilator such as the increase of the peak inspiratory pressure, the air-trapping, the drop of the volume, and the asynchrony between the machine and the patient. In our scenario, we are going to focus on air-trapping, in which air is 'trapped' in the lungs during expiration [7], as well as the asynchrony problem mentioned above, which is separated in four conditions, the trigger breath, the flow hunger, a situation, in which the patient wants to expire before the ventilator is planned to do so and the volume hunger [7].

3 DR SUPER VR SIMULATION

The experience is developed using Unity3D game engine and the main hardware equipment used is the Oculus Rift HMD. Due to the need of a realistic VR simulation, a wide range of medical 3d objects, human organs, muscles, cavities were either imported or modelled by using the Blender3D software.

Scenario: In the beginning, the user is being placed into a main menu scene, to select the desired mode. In both tutorials, specific guidelines regarding the intubation and ventilator troubleshooting procedures are aligned with certified sources. To help the inexperienced users (regarding the VR or/and the medical procedure) to complete these challenges, certain checkpoints, for each action, are present. Constructive feedback is also provided, so the players are capable of reviewing their actions, before entering the simulated ICU experience. During the main game, these steps must be implemented without any assistance, to ensure that the user's skills are acceptably proficient. Subsequently, the player is placed in an Emergency Room – ICU, where virtual patients are either suffering from ventilator complications or new ones that need to be intubated are inserted into the room. The players have at their disposal a limited amount of lives/error opportunities, which they are able to regain, in case of fallaciously implementing a procedure, by staying alive for more time in the game. The frequency of complications increases while the time passes to further cultivate the desired skills.



Figure 2: Patient's monitor which visualises the graphs of the air flow and the lungs' volume.

Multiplayer interactions: By enabling the multiplayer capability in our simulation we "force" two end-users to be the doctors/medical staff in the same ICU. To elaborate, during our game, we choose to alert the players for the new complications by using haptic feedback, which is presented later, and a monitor inside the VR simulation. In the multiplayer mode, the increasing rate of the incoming patients/complications along with the fact that the players have as a team, 3 lives/chances of mistake, highlight the importance of fruitful communication between the members. Our simulation can also be perceived as an assistance tool for medical training, due to multiplayer voice communication, thus enabling individuals to learn more while having fun.

Haptic device: Intubation is a delicate procedure, so to simulate it realistically, a 3D geomatic haptic touch device is also employed. This provides a force feedback, which is essential during the penetration of the laryngoscope in the oral cavity. Moreover, it promises greater control, flexibility, and touch manipulation, vital features for instructing intubation methods.

Vibrator motor: Two distinct vibrator patterns (for intubation procedure and ventilator troubleshooting respectively) are used to alert the user for potential patients' complications. For that reason, a vibrator motor is driven by an Arduino with a Bluetooth module attached as shown in figure 3. The circuit is stored in a black case and placed in the breast pocket. Until an individual complication is addressed the vibration frequency is increasing with time to alert the user. When a player reaches to the specific destination, the vibration stops.

Concerning the user experience, we conducted a research by interviewing 11 participants (5 F-6 M), including postgraduate and undergraduate students from the University of Patras, who either had or did not have previous experience in VR applications. After completing a questionnaire with demographic information, they were given a short presentation of the game, they proceeded in the training session and then the main game. The results indicated that users with VR experience were more effective than the other group.

4 CONCLUSION

Regarding the 3DUI 2021 contest objectives, this paper presents a virtual reality 3DUI interaction system that serves as the medium to familiarize the end-users with the main intubation procedures and ventilator troubleshooting actions. The principal technical contributions are the use of an interactive scenario, simulating the intubation procedure and possible complications with training purposes, in first person, as well as a 3D Systemic haptic device to provide resistance feedback to the user and improve the sense of embodiment. One of the major challenges was the design process of a realistic inner human body representation and overall interactions. We strongly believe that these implementations could have a significant impact on the rapid interactive training of medical personnel during the COVID-19 pandemic, thus addressing a wide range of health education related applications.

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